ENVS193DS Final

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Table of contents

GitHub repo

Access my repo here.

Set up

```
# read in packages
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(DHARMa) # for diagnostics
library(ggeffects) # for ggpredict
library(flextable) # for flextable
library(gtsummary)

# read in data
nests <- read_csv(here("data", "nest_data_final.csv"))
# read in personal data to object from csv
my_data <- read_csv(here("data", "envs193ds-personal-data.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

b. Suggestions for rewriting

c. Further analysis

Problem 2. Data analysis

*DO NOT include any code or output from your data exploration

a. Clean and wrangle

```r
# cleaned data
nests_clean <- nests |>
  clean_names()
```

b. Response variable

c. Purpose of study

d. Table of models

e. Run the models

f. Select the best model

g. Check the diagnostics

h. Visualize the model predictions

i. Write the caption for your figure

j. Calculate model predictions

k. Interpret your results

Problem 3. Affective and exploratory visualizations

a. Comparising visualizations

b. Designing an analysis

c. Check any assumptions and run your analysis

Setting up

```
# cleaned column names
my_data <- my_data |>
  clean_names()

# created new column: 1 if workout was a run, 0 if not
my_data <- my_data |>
  mutate(is_run = case_when(
    workout_type == "Run" ~ 1, # run = 1
    .default = 0))              # all other workout types = 0
```

Check assumptions

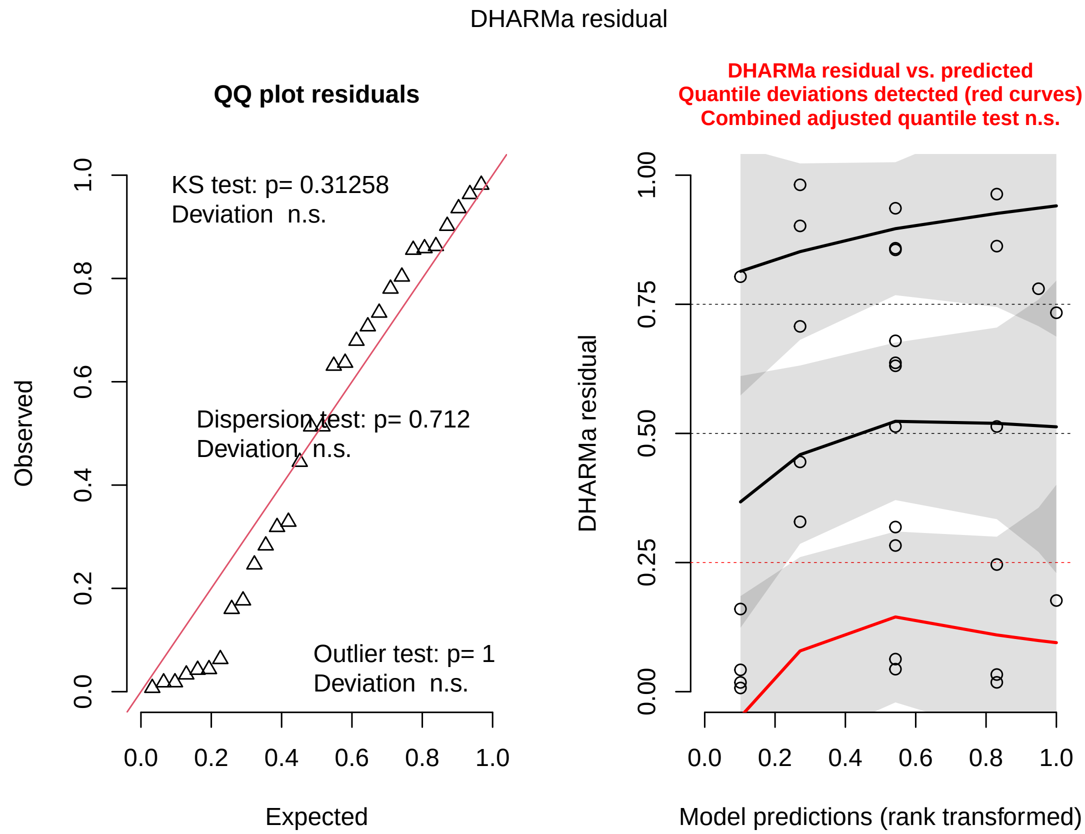
NEED TO DO indep obs, what else...

Run the analysis from part b

```r
# fitted generalized linear model for running
# response: whether or not workout was a run (binary: 1/0)
# predictor: sleep the previous night (hours)
run_mod <- glm(
  # formula: binary run outcome ~ sleep hours
  is_run ~ sleep_hours,
  # data: my_data data frame
  data = my_data,
  # binomial error distribution for binary response
  family = binomial)
```

Checking diagnostics

```r
# simulated residuals from GLM for diagnostic plots
plot(simulateResiduals(run_mod))
```



Info from left plot (QQ plot of residuals):

- KS test: do my residuals follow a uniform distribution? → generally yes (CHECK)
- Dispersion test: are your residuals more dispersed than expected from a uniform distribution? → no
- Outlier test: are there outliers in your data set that could be influencing model coefficients or predictions? → no

Info from right plot (residuals vs predicted values):

- By default, R generates the reference lines for quartiles: 25, 50, 75
- Not that informative but look at patterns within residuals across the range → DHARMa diagnostics flag issues in 25th quartile due to small sample size, not because the model is wrong.

Looking at the model coefficients

```r
# displayed model summary
summary(run_mod)
```

```
Call:
glm(formula = is_run ~ sleep_hours, family = binomial, data = my_data)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -8.7011     3.4964  -2.489   0.0128 *
sleep_hours   1.0873     0.4712   2.307   0.0210 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 36.652  on 29  degrees of freedom
Residual deviance: 28.046  on 28  degrees of freedom
AIC: 32.046

Number of Fisher Scoring iterations: 5
```

Calculate inverse logit

```r
# calculated inverse logit
plogis(-8.7011)
```

```
[1] 0.000166375
```

Note:

- inverse logit is 0.00016
- The probability of going on a run if sleep duration is 0 hours is 0.00016?

Calculate an effect size

```r
# calculated odds ratio
gtsummary::tbl_regression(run_mod,
                          exponentiate = TRUE)
```

Characteristic	OR	95% CI	p-value
sleep_hours	2.97	1.37, 9.40	0.021

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Note:

- <u>results at top of this page</u>
- odds ratio is 2.97
- 2.97 > 1 so the probability of going on a run increases by a factor of 2.97 with every 1 hour increase in sleep

Summary: For every 1 hour increase in sleep, the the odds of going on a run increase by a factor of 2.97 (95% CI: [1.37, 9.40], p = 0.021,   = 0.05).

**Looking at the model predictions (where to put this section?)

- using probabilities to determine chance of getting certain outcome
- line represents the probability of 1 (going on a run) actually occurring

```r
# what is the probabilty of going on a run after sleeping for 10 hours?
ggpredict(run_mod,
          terms = "sleep_hours [10]")
```

```
# Predicted probabilities of is_run

sleep_hours | Predicted |     95% CI
------------------------------------
         10 |      0.90 | 0.39, 0.99
```

```r
# what is the probabilty of going on a run after sleeping for 4 hours?
ggpredict(run_mod,
          terms = "sleep_hours [4]")
```

```
# Predicted probabilities of is_run

sleep_hours | Predicted |     95% CI
------------------------------------
          4 |      0.01 | 0.00, 0.25
```

Note:

- If I slept for 10 hours, the probability of going on a run the next day is 0.90 (95% CI [0.39, 0.99]).
- If I slept for 4 hours, the probability of going on a run the next day is 0.01 (95% CI [0.00, 0.25]).

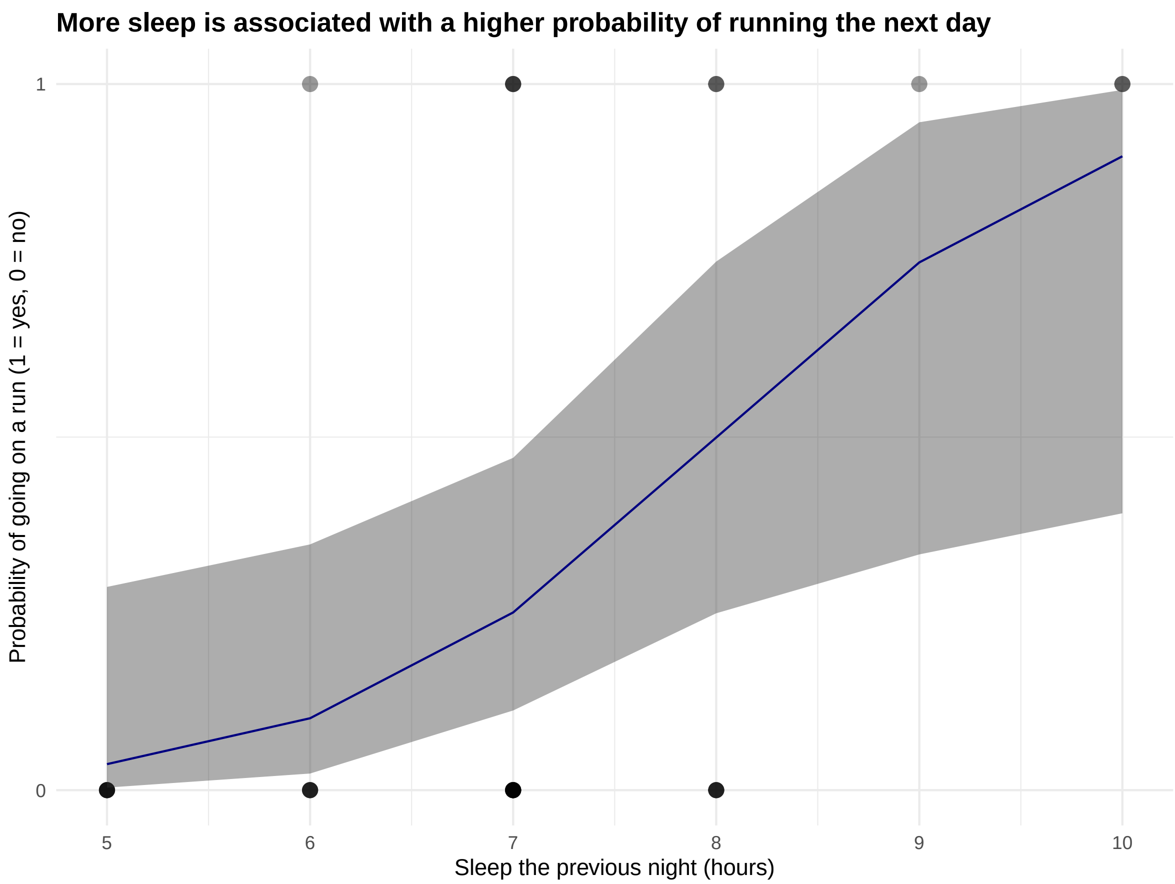
d. Create a visualization

Visualizing model predictions

```
# generated model predictions across range of sleep values in data set
mod_preds <- ggpredict(run_mod,
                       terms = "sleep_hours [5:10 by = 1]")

# base layer: ggplot with raw data
# data frame: my_data
ggplot(my_data,
       aes(x = sleep_hours,
           y = is_run)) +
  # first layer: raw observations
  geom_point(size = 3,
             alpha = 0.4) +
  # second layer: confidence interval ribbon from model predictions
  geom_ribbon(data = mod_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
                  alpha = 0.4) +
  # third layer: logistic curve from model predictions
  geom_line(data = mod_preds,
            aes(x = x,
                y = predicted),
            # custom color
            color = "navyblue") +
  # constrained y-axis to 0 and 1
  scale_y_continuous(limits = c(0, 1),
                     breaks = c(0, 1)) +
  # relabeled axes and added title
  labs(title =
"More sleep is associated with a higher probability of running the next day",
       x     = "Sleep the previous night (hours)",
```

```
    y      = "Probability of going on a run (1 = yes, 0 = no)") +
  # changed theme from default
  theme_minimal() +
  # bolded title
  theme(plot.title = element_text(face = "bold"))
```



Describing the model

```
# displaying model in table
as_flextable(run_mod)
```

	Estimate	Standard Error	z value	Pr(>lzl)	
(Intercept)	-8.701	3.496	-2.489	0.0128	*
sleep_hours	1.087	0.471	2.307	0.0210	*

*Signif. codes: 0 <= ’***’ < 0.001 < ’**’ < 0.01 < ’*’ < 0.05*

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.65 on 29 degrees of freedom

Residual deviance: 28.05 on 28 degrees of freedom

e. Write a caption

Figure caption

f. Write about your results

ADD For every 1 hour increase in sleep, the the odds of going on a run increase by a factor of 2.97 (95% CI: [1.37, 9.40], $p = 0.021$, $= 0.05$).

g. Sharing your affective visualization

Presented visualization in Week 10 Workshop.